

**Contribution to the HLPE-FSN Online Consultation on
"Harnessing Artificial Intelligence, Digitalization and Data
Governance for Food Security and Nutrition"**

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**Are the issues identified by the HLPE-FSN the most important issues
related to Artificial Intelligence, digitalization and data governance
affecting food security and nutrition (FSN), globally and in specific
contexts?**

The issues identified are largely accurate and the note's insistence on
evidence over vendor claims is very valuable. However, the framing
consistently mistakes a technical implementation gap for a governance
gap. Data sovereignty, fair benefit-sharing, and "contestability" are
not primarily waiting for better policy language. They are waiting for
technical infrastructure that can operationalize that language. The
note at its core reads as a well-intentioned charter – it will be cited
and applauded, and then it will produce little to no change, because
the people who could implement it cannot extract actionable technical
specifications from it, and the people who wrote it may not have
engaged with what those specifications would look like.

Are there any other key issues that should be added and elaborated?

I see you have used the word "dataspaces" but you do not engage with what building one will actually require: machine-readable policy representations, ontological frameworks for rights and obligations, interoperability standards, and provenance mechanisms for data and model lineage. There could be very real things that could be done related to food crisis, food shortage, food quality, and so on, because generic one-size-fits-all solutions are not good for domain-specific problems. A lot of work on autonomous vehicles exists because big companies want money. Where is the work on food? The existing European data space framework – the Data Act, Data Governance Act, European Health Data Space regulation – each have measurable gaps between their stated governance principles and what is technically implemented (or even implementable) today. Without bridging that gap, recommendations to "implement dataspaces" remain aspirational. The data spaces in Europe have been lagging forever. What is needed, I have reasons to believe, is sustained investment in policy-aware informatics, infrastructure, and research, not more policy coordination. Especially in fields like this, we could save lives if technology was adapted to specific needs.

Are the issues identified fully capturing the links with FSN outcomes?

The note's FSN outcome metrics (HDDs, FIEs, HFIAAs) are appropriate. But their usefulness depends on data pipelines that do not exist in most high-need contexts. Measuring the FSN impact of digital tools requires the same data infrastructure that those tools are supposed to build – a circularity the note does not acknowledge.

Is there any aspect of direct or indirect FSN outcome that should be further elaborated?

Labor conditions under which AI training data is produced deserve attention. Small populations whose agricultural data is extracted for model training are often also the source of low-paid data annotation labor, creating a double extraction relation with direct implication for rural livelihood and food security that this note does not address.

In particular, do you have examples of effective policies to improve FSN outcomes of the use of Artificial Intelligence, digitalization and data governance in food systems?

Work on extracting machine-readable ODRL policies from business process models (<https://bpmn.linkeddata.es/>) and on building open linked data portals for benchmarking AI agents in health data spaces (<https://mcp.linkeddata.es/>) will show you that the technical path from policy aspiration to implementation can exist and is very easy. Both cases show that the bottleneck is not technical incapability. It is the absence of institutional structures that bring legal informatics researchers, domain practitioners, and policymakers into sustained collaboration – rather than these one-off consultations.

Is there any missing reference to key literature and data?

The European Health Data Space can provide a concrete and instructive test case for food data governance. It is a well-resourced, politically supported framework with a significant gap between its stated data sovereignty principles and its current technical implementation. If you want to close that gap, it is not better regulation but better translation infrastructure that will help: interoperable data portals and so on. The same logic applies to FSN dataspace, like in contexts where smallholder data is collected by private platforms under terms that neither farmers nor regulators can technically audit or contest.

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